

Charles University
Faculty of Social Sciences
Institute of Economic Studies



HANDBOOK

Financial Markets Instruments I
JEM035

Winter Semester
Academic Year: 2026/2027

Last update: September 29, 2026

1. Course schedule

The course consists of lectures and seminars – there is one lecture and one seminar taught each week.

- Lectures are on Tuesdays: 14:00 – 15:20 in room number 314.
- Seminars are on Tuesdays: 15:30 – 16:50 in room number 109.

You are expected to be present in person for both lectures and seminars. The attendance is not mandatory but recommended.

2. Contact & communication

For any course related questions please contact prof. Dědek (dedek@fsv.cuni.cz) or teaching assistant Petr Polák (petr.polak@fsv.cuni.cz). **Mainly, communicate with us in person!**

Please kindly note that e-mails sent to different e-mail addresses than the ones stated above will not be taken into account.

Questions related to lectures should be addressed to prof. Dědek, questions about seminars and exams to the TA. Use seminars to ask your questions, since other might benefit from it as well.

Please always use specific subject of the email **"FMI I:"** and then continue with your problem. **Example:** *"FMI I: Written test"* or *"FMI I: Seminar 5"*

3. Course contents

The course content is divided into two sections – Futures and Options.

Part I: Futures Contracts

1. Basic features of forward and futures trading: distinction between forward and futures contracts, practical arrangements (clearing house, operation of margins, marking to market, delivery process, trading limits), basis (contango, backwardation).
2. Examples of financial futures contracts: stock index futures, currency futures, short-term interest rate futures, long-term interest rate futures (price factor, CTD bond).
3. Model cost-of-carry: derivation of basic formula, valuation of futures contracts, fair pricing of futures contracts (covered and uncovered interest rate parity, implied forward rates).
4. Speculative trading strategies with futures: features of speculative trade (working of leverage), open position trading (long and short positions), spread trading (intra-contract spread, butterfly spread), basis trading.

5. Arbitrage trading strategies with futures: features of arbitrage trade, box arbitrage, conversion arbitrage. Hedging trading strategies with futures: typology of hedging, hedging using spot transactions, basic elements of futures hedging (hedge ratio, hedge efficiency), basis risk, interpolative hedge, minimum variance hedge ratio, practical examples of hedging with futures.

Part II: Option Contracts

6. Basic features of option contracts: definition of call and put options, profit and loss profiles, decomposition of option premium, margining in option contracts.
7. Examples of financial option contracts: equity options, stock index options, currency options, options on interest rate futures contracts.
8. Option combinations: synthetic security, combinations (straddle, strangle, strap, strip), spreads (vertical and horizontal, synthetic futures, butterfly, condor), combinations with the underlying asset.
9. Fair pricing of options contracts: binomial model, Black-Scholes model, volatility smile, sensitivity analysis, delta-hedging, gamma-hedging, put-call parity.
10. Trading strategies with options: speculative option strategies (open position trading, spread trading, volatility trading), arbitrage option strategies (credit and debit box arbitrage, conversion arbitrage), hedging option strategies (protective put, covered call, fixed and delta-neutral hedge, portfolio insurance, synthetic put option).
11. Exotic options: engineering with the payoff profile (Asian options, package options, binary options, look-back options), engineering with the content of financial instrument (barrier options, chooser options), engineering with the underlying asset (compound options, multiple-asset options).

4. Grading

The grading of the course consists of two parts:

- **20% Activity points – students can obtain them mainly during seminars**
- Seminar points can be also gained by extra activity such as solving prof. Dėdek's examples – more info during lectures.
- **80% Written exam – student must receive at least 50% of the final exam to be eligible for passing the course.**

In total, a student can obtain 80 points from the written examination and 20 points from the seminars (100 points is the total maximum).

A student has three attempts to pass the course. That means that if a student does not receive at least 50 percent from the first written test, he/she still has a chance to write

¹Sum of points from written examination and points received from presentation.

Final Score ¹	Grade
0 - 50	F
51 - 60	E
61 - 70	D
71 - 80	C
81 - 90	B
91 - 100	A

the test and receive a higher score. It is also NOT possible to improve score in written exam once student has passed the minimum requirement. A student has to reach at least 50 points in total to pass the course (getting 45 points from exam and zero from seminars is also marked as failed.)

5. Seminars

Each seminar will be devoted to short recap of the theory and mainly to solving exercises.

In the first part of the seminar, the TA will briefly summarize the current topic and then solve about three exercises on board. The aim is to get you familiar with the topic and show how related problem are solved.

In the second (and much shorter) part of the seminars, you have a possibility to get **seminar points**. You will be provided with an exercise and about 15-20 minutes to solve it. Those, who will manage will get points. You can get 3 points at each seminar. In total, you can get up to 20 seminar points, hence you do not have to be present or able to solve the exercise on every seminar to get the maximum points. You are expected to bring the list of formulas with you and a calculator so you can solve the exercises in class.

The setting of the exercises solved at seminars will be provided to you in SIS, but the solution will not.

Feel free to ask any questions related to the organizational structure of the course in person during seminars to avoid possible misunderstanding.

6. Exam

There will be only written exam which will consist of two parts – **theoretical questions** (multiple choice and open questions) and **exercises** (based on the exercises from seminars, lectures and Prof. Dedek's webpage). The exam will be rewarded by 80 points. A student will receive full amount of points only if the solution will be presented carefully with detailed description of all the necessary steps. Theory part is worth 40 points, then 4 exercises worth 10 points each.

During the written exam:

- students are allowed to use only a non-programmable calculator

- it is **strictly prohibited** to use electronic communication devices such as smart-phones or laptops
- it is **strictly prohibited** to use books, lecture notes, own notes etc.
- teamwork and/or the exchange of notes is strictly prohibited
- students will be provided with the list of formulas, which is available in SIS for your information in advance.

7. Examination planning

There will be three dates to write the test in the examination period so that each student can exploit all three chances to pass the course. It is thus not advisable to wait until the last examination date.

There will be also one exam during last seminar – all students can attend this exam. The dates of the written examination are following, times and rooms are subject to possible change:

Date	Time	Room
<i>15.12.2026</i>	<i>15:30-16:50</i>	<i>O314</i>
5. 1. 2027	15:30-16:50	?
19. 1. 2027	15:30-16:50	?

The SIS registration for the final written examination is compulsory.

8. Studying materials & Literature

There are several types of studying materials available for you. The most important are the presentations (available in SIS), but there are also additional materials available on the web page of prof. Dědek (dedeklegacy.cz).

- Handouts: 01 Futures Contracts, 02 Option Contracts
- Blake, D. (2000) Financial Market Analysis(second edition), McGraw-Hill.
- Hull, John C. (2006) Options, Futures and Other Derivatives, 6th edition, Prentice Hall, New Jersey
- Bjork, T (2009) Arbitrage Theory in Continuous Time. 3rd edition. Oxford.
- Jílek, Josef (1995) Termínové a opční obchody, 1. vydání, Grada Publishing, Praha
- Jílek, Josef (2005) Finanční a komoditní deriváty v praxi, Grada Publishing, Praha
- Black, F. and Scholes, M. (1973) The pricing of options and corporate liabilities. Journal of Political Economy 81, 637–659.